READING PASSAGE 2

*You should spend about 20 minutes on **Questions 14–26**, which are based on Reading Passage 2 below.*

War of the Plants

Professor John Lovett from New England University, Australia, describes some of the ways in which plants ensure their own survival.

A The general perception of plants is that they are defenceless when faced with threat from animals (including man), insects and other plants. This, however, is not the case. Plants are able to defend themselves against attack from their environment in many ways. There are two major features of plants that ensure their continuation as species. One of these relates to their physical attributes. Spinifex grass, a plant native to Australia, provides a notable example. Adapted to life in the hot, desert inland, spinifex has developed tightly rolled leaves which reduce the effects of drought stress. The hairs, or trichomes, which are found on the inner surface of the rolled-up spinifex leaf are believed to assist in minimising water loss. Trichomes also have defensive roles against predators. Many plants, such as the sunflower, have hairy leaves which are rough to the touch. When the plant is bitten by a grazing animal, the sensation is unpleasant on the tongue and the animal normally moves on to find more palatable food.

B Some families of plants such as the Solanaceae, which includes the tomato, the potato and other food plants, feature glandular trichomes. In these structures a tiny sac is carried on the hair, which rises from the epidermis, or skin, of the plant. The sac contains chemicals which may perform a range of functions. For example, an insect landing on a tomato leaf and rupturing the sac is exposed to a toxic chemical which will deter it from feeding on that plant in future. In other plants of the same family, the sacs contain a natural adhesive and the unsuspecting insect landing on such a leaf and rupturing a trichome sac is held onto the plant, unable to depart. In the leaves of the nettle plant these trichomes resemble miniature hypodermic syringes. An irritant chemical is stored at the base of the ‘syringe’. If an animal touches it, a shot of the chemical is discharged through the ‘needle’ and into the skin of the animal. The irritation the chemical causes is a deterrent against future threat.

C The poisons and glues of the Solanaceae and the irritant of the nettle indicate the other major method of plant survival: that some plants use chemicals to protect themselves, sometimes from each other. Chemical interactions that occur between plants are known as ‘allelopathy’, and in natural plant communities it is one of many factors which determine how plants grow in relation to one another. An example in Australia is the suppression of vegetation growth beneath eucalypt trees, an effect which decreases with increasing distance from the trunk of the tree.

D The most detailed studies of allelopathy have generally been carried out on agricultural plant communities. In agriculture, allelopathy has been identified particularly with weeds, which compete with desired crop or pasture species, such as wheat, oats and grass, for environmental resources. Competition can be physical, where the growth of one plant maximises the use of water, nutrients and light to the disadvantage of another. Allelopathy, on the other hand, involves plant-produced chemicals which may poison a neighbouring plant. On the whole, weeds are far more physically and chemically aggressive than crop and pasture species. As an example, recent research on the common thorn apple, Datura stramonium, a weed found throughout Australia, has found that two powerful chemicals are released from the seed during the early stages of germination. In fact, it is necessary for the chemicals to be released from the seed coat so that the seed can germinate. Once released, the chemicals then provide a barrier that inhibits the growth of any potential plant competitor near the thorn apple seedling.

E Allelopathic chemicals function in quite subtle ways. Generally they are active in small quantities, they are effective only with certain other plants, they act by disrupting the plant's natural processes and they have no known residual effects. It has also been found that the activity of these chemicals may be synergistic, that is, two or more such substances may combine to produce an effect that is greater than the effects of the substances acting in isolation. These are the very attributes which are being targeted in the current development of synthetic *herbicides**. Allelopathy is not restricted to weeds in agriculture, but allelopathic activity in crop and pasture plants appears to have diminished, possibly because plant breeders may have concentrated on improving aspects of crop quality and quantity rather than aspects of defence against natural enemies.

F Allelopathic chemicals may be harvested for development as 'natural herbicides', similar to the harvest of pyrethrins from the pyrethrum daisy for use as natural insecticides. Harnessing allelopathy is highly attractive to the farming industry, as at least half of the more than A$200-million annual cost of crop protection in Australia is spent on herbicides. Enhancing allelopathic activity in crop and pasture plants could reduce this enormous bill. Developing and harvesting natural herbicides might also offer real advantages in addressing environmental concerns about the over-use of synthetic chemicals currently used in crop protection. In many parts of the world, communities are showing an increased willingness to pay a premium for products grown with few or no synthetic chemicals, and natural products are becoming increasingly attractive.

**herbicides: chemicals that kill plants, used mainly to control weeds*

Questions 14–21

Reading Passage 2 has six paragraphs, ***A–F****.*

Which paragraph contains the following information?

Write the correct letter, ***A–F****, in boxes 14–21 on your answer sheet.*

NB *You may use any letter more than once.*

14 an example of a plant's use of chemicals in its reproductive process

15 how some plants use a kind of glue to discourage attack

16 two examples of non-chemical methods of defence in plants

17 a definition of the term allelopathy

18 descriptions of how plants use chemicals to discourage predators

19 benefits of using plant-based chemicals to control weeds in agricultural areas

20 a description of the characteristics of allelopathic chemicals

21 an example of a physical characteristic that helps one plant conserve water

Questions 22–25

Do the following statements agree with the information given in Reading Passage 2?

In boxes 22–25 on your answer sheet, write

TRUE	*if the statement agrees with the information*
FALSE	*if the statement contradicts the information*
NOT GIVEN	*if there is no information on this*

22 The 'hypodermic syringes' of the nettle plant are an example of glandular trichomes.

23 Allelopathic plants grow more successfully than plants that rely only on physical features.

24 New research into synthetic herbicides is largely focused on individual allelopathic chemicals.

25 Herbicides cost Australian farmers over A$200 million a year.

Question 26

Choose the correct letter, ***A, B, C*** *or* ***D****.*

Write the correct letter in box 26 on your answer sheet.

The two main points of this article are that plants can defend themselves in a number of ways, and that

A combating weeds is a major problem for Australian farmers.
B considerable research has been carried out into the use of plant chemicals.
C Australian plants have unique methods of self-protection.
D chemicals found in weeds can have practical benefits for agriculture.

Disclaimer
Compiled, formatted, and lightly proofread by ZYZ Reading Walks.
All copyright in the underlying works remains with the original authors and publishers.
No affiliation with or endorsement by any rights holder (including IELTS® owners).
For non-commercial educational use only. This notice must remain intact in all copies.
Available free of charge from ZYZ Reading Walks. Resale or any paid distribution is prohibited.

Questions 14–21 段落信息匹配题

题号	答案	题干翻译	详细定位与翻译	详细解释
14	D	一种植物在繁殖过程中使用化学物质的例子	**第D段**：recent research on the common thorn apple… has found that two powerful chemicals are released from the seed during the early stages of germination. In fact, it is necessary for the chemicals to be released from the seed coat so that the seed can germinate. 译：最近对曼陀罗这种杂草的研究发现，在发芽早期，种子会释放两种强效化学物质。事实上，这些化学物质必须从种皮中释放出来，种子才能发芽。	题干关键词是 **reproductive process**，这里对应 **germination 发芽**，属于植物繁殖/生长初期过程；**use of chemicals** 对应 two powerful chemicals are released。
15	B	一些植物如何使用一种胶状物来阻止攻击	**第B段**：In other plants of the same family, the sacs contain a natural adhesive and the unsuspecting insect… is held onto the plant, unable to depart. 译：在同一科的其他植物中，这些小囊含有一种天然黏合剂，毫无防备的昆虫落在叶片上并弄破毛囊后，就会被黏在植物上，无法离开。	题干中的 **a kind of glue** 对应原文 **a natural adhesive**；**discourage attack** 对应昆虫被黏住后无法继续取食或攻击。
16	A	两个植物非化学防御方法的例子	**第A段**：One of these relates to their physical attributes… spinifex has developed tightly rolled leaves… Trichomes also have defensive roles against predators. Many plants… have hairy leaves which are rough to the touch. 译：其中一种防御方式与植物的物理特征有关。刺草发展出紧紧卷起的叶子……毛状体也有抵御捕食者的作用。许多植物有带毛的叶子，触感粗糙。	题干重点是 **non-chemical methods**，即非化学方法。第A段讲的是 **physical attributes 物理特征**，包括卷叶减少干旱压力、毛状叶片让动物口感不适，属于两个物理防御例子。
17	C	“allelopathy” 这个术语的定义	**第C段**：Chemical interactions that occur between plants are known as ‘allelopathy’. 译：植物之间发生的化学相互作用被称为 “化感作用”。	这是非常直接的定义句。题干中的 **definition of the term allelopathy** 对应原文 **are known as ‘allelopathy’**。
18	B	植物如何使用化学物质来阻止捕食者的描述	**第B段**：an insect… is exposed to a toxic chemical which will deter it from feeding… / An irritant chemical is stored at the base of the ‘syringe’… The irritation… causes is a deterrent against future threat. 译：昆虫接触到一种有毒化学物质，以后就不再取食该植物……一种刺激性化学物质储存在 “注射器” 底部，动物触碰后，化学物质会被注入皮肤，造成刺激，从而阻止未来的威胁。	第B段集中描述植物用 **toxic chemical 有毒化学物质** 和 **irritant chemical 刺激性化学物质** 来对付昆虫或动物，因此对应 “用化学物质阻止捕食者”。
19	F	在农业区域中，使用植物来源化学物质控制杂草的好处	**第F段**：Enhancing allelopathic activity in crop and pasture plants could reduce this enormous bill. Developing and harvesting natural herbicides might also offer real advantages in addressing environmental concerns about the over-use of synthetic chemicals… 译：增强农作物和牧草中的化感作用可以减少这笔巨大开支。开发和收获天然除草剂也可能在应对合成化学品过度使用所带来的环境担忧方面具有真正优势。	题干中的 **benefits** 对应第F段的两个好处：一是降低农业除草成本，二是减少对合成化学品的依赖，更环保。

20	E	化感化学物质特征的描述	**第E段**：Generally they are active in small quantities, they are effective only with certain other plants, they act by disrupting the plant's natural processes and they have no known residual effects... may be synergistic... 译：一般来说，它们在很小剂量下就能起作用，只对某些其他植物有效，通过干扰植物的自然过程发挥作用，并且没有已知残留影响……这些化学物质还可能具有协同作用。	题干中的 **characteristics** 对应第E段列举的一系列属性：小剂量有效、选择性作用、干扰自然过程、无残留、可协同。
21	A	一种帮助植物保存水分的物理特征例子	**第A段**：spinifex has developed tightly rolled leaves which reduce the effects of drought stress. The hairs, or trichomes... are believed to assist in minimising water loss. 译：刺草发展出紧紧卷起的叶子，可以减轻干旱压力。叶子内侧的毛状体被认为有助于减少水分流失。	题干关键词是 **physical characteristic** 和 **conserve water**。第A段中的 **tightly rolled leaves** 和 **trichomes minimising water loss** 都是帮助保存水分的物理特征。

Questions 22–25 TRUE / FALSE / NOT GIVEN

题号	答案	题干翻译	详细定位与翻译	详细解释
22	TRUE	蓖麻植物的“皮下注射器”是腺毛的一种例子。	**第B段**：Some families of plants... feature glandular trichomes. In these structures a tiny sac is carried on the hair... In the leaves of the nettle plant these trichomes resemble miniature hypodermic syringes. 译：一些植物科具有腺毛。在这些结构中，一个小囊长在植物表皮伸出的毛上……在荨麻叶片中，这些毛状体像微型皮下注射器。	原文先介绍 **glandular trichomes 腺毛**，然后说荨麻中的 **these trichomes** 像微型注射器。题干说荨麻的“hypodermic syringes”是glandular trichomes 的例子，与原文一致。
23	NOT GIVEN	具有化感作用的植物比只依靠物理特征的植物生长得更成功。	相关信息分散在**第A段**和**第C-D段**。第A段讲植物的物理防御；第C-D段讲化感作用和植物间化学竞争。但原文没有比较“化感植物”和“只依靠物理特征的植物”谁生长得更成功。	题干是一个比较判断：**grow more successfully than...**。文章只说植物有两大防御方式，也说化感作用会影响其他植物生长，但没有说化感植物整体上比只用物理防御的植物更成功，所以是 NOT GIVEN。
24	FALSE	关于合成除草剂的新研究主要集中在单个化感化学物质上。	**第E段**：It has also been found that the activity of these chemicals may be synergistic, that is, two or more such substances may combine to produce an effect that is greater than the effects of the substances acting in isolation. These are the very attributes which are being targeted in the current development of synthetic herbicides. 译：研究还发现，这些化学物质的活性可能具有协同作用，也就是说，两个或多个此类物质结合起来产生的效果可能大于它们单独作用时的效果。这些正是当前合成除草剂开发所瞄准的特征。	题干说研究主要关注 **individual allelopathic chemicals 单个化感化学物质**，但原文强调的是 **two or more substances may combine 协同作用**，即多个物质组合产生更强效果。题干与原文方向相反，所以是 FALSE。
25	FALSE	除草剂每年让澳大利亚农民花费超过2亿澳元。	**第F段**：at least half of the more than A$200-million annual cost of crop protection in Australia is spent on herbicides. 译：澳大利亚每年超过2亿澳元的作物保护成本中，至少有一半花在除草剂上。	原文说的是 **crop protection 作物保护总成本超过2亿澳元**，其中至少一半用于 herbicides。也就是说除草剂成本至少超过1亿澳元，但不能推出“除草剂本身超过2亿澳元”。题干把总成本误读成除草剂成本，因此 FALSE。

Question 26 单选题

题号	答案	题干翻译	详细定位与翻译	详细解释
26	D	本文的两个主要观点是：植物能够用多种方式保护自己，并且……	**第 A 段**：Plants are able to defend themselves against attack… in many ways. 译：植物能够以多种方式保护自己免受攻击。**第 F 段**：Developing and harvesting natural herbicides might also offer real advantages… 译：开发和收获天然除草剂也可能带来实际优势。	文章前半部分说明植物如何自我防御，包括物理方式和化学方式；后半部分重点转向化感化学物质在农业中的潜在应用，尤其是天然除草剂。因此第二个主旨应是 **chemicals found in weeds can have practical benefits for agriculture**。
26	排除 A	与杂草作斗争是澳大利亚农民的主要问题。	第 D-F 段确实提到 weeds 和农业成本，但文章重点不是“杂草问题很严重”，而是植物化学物质的应用价值。	A 只抓住农业背景中的一个局部信息，不是全文第二主旨。
26	排除 B	大量研究已经针对植物化学物质的使用展开。	第 D 段说最详细的化感研究通常在农业植物群落中进行，但这不是全文核心结论。	B 偏向“研究情况”，而文章最后强调的是实际应用和农业价值。
26	排除 C	澳大利亚植物有独特的自我保护方法。	文中确实举了澳大利亚植物例子，如 spinifex、eucalypt、thorn apple，但并没有强调“澳大利亚植物独有”。	C 的 **unique 独特的** 没有原文依据，且范围过窄。
26	正确 D	杂草中的化学物质可以给农业带来实际好处。	第 F 段：Allelopathic chemicals may be harvested for development as ‘natural herbicides’… Enhancing allelopathic activity… could reduce this enormous bill. 译：化感化学物质可以被收获并开发成天然除草剂……增强农作物和牧草中的化感活性可以减少巨额开支。	D 准确概括了文章后半部分：虽然 weeds 是农业中的竞争者，但其中的化学机制可以被利用，发展天然除草剂，降低成本并回应环保担忧。